

Repair Request Test Case Specification

System/UAT + Performance Verification Extension

Document Control	Value
Document ID	RR-TC-001
Version	1.3 - Performance Revision Candidate
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Supersedes	RR-TC-001 v1.2
Revision Scope	Add repeatable performance test design, evidence and regression protection while keeping functional expected states/audits unchanged.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

Repair Request Test Case Specification

System Test + UAT / MVP Baseline Revision

Document Control	Value
Document ID	RR-TC-001
Version	1.2 - Test Design Baseline Candidate
Status	Content Complete - Ready for Test Design Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	FR-01..09; BR-01..19; RR-STS-001 v1.5; UC v1.6; RR-DD-001 v1.3
Audience	QA Lead, Tester, Business Owner, SA, Developer, Security Reviewer
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	กำหนด System Test, Security, Concurrency, Notification, NFR and UAT acceptance with deterministic expected state/audit.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. Test Strategy / Execution Rule

- Test allowed UI/API actions only; direct state update is not a valid normal-path test.
- Each execution records Actual Result, test data, evidence/log/audit/correlation, tester, date, build and defect ID.
- Critical/High UAT=Yes must pass; Critical/High open defects = 0 before sign-off.
- Controlled clock is required for SLA Risk T-4h, breach and override behavior.
- At least two tenant/site scopes are required for IDOR/cross-scope denial.

2. Common Test Data

Area	Required
Roles	Requester, Approver, Coordinator, Technician, Team Lead, Supervisor, designated Requester/Site Contact, Administrator, unauthorized user
Scope	At least 2 tenant/site scopes
Masters	Active/inactive Site, Equipment, Location, Category, Priority, Outcome, Evidence Type, Team, Approval Route
Files	CLEAN valid files, invalid type, >10MB, PENDING, FAILED
Clock	SLA +24h, risk T-4h, breach, override, timestamp order
Concurrency	Two clients with same current token
Performance	100 concurrent users, 100k records/Site, representative indexed filters/sort/paging

3. System Test Cases

ID	Pri.	Traceability	Scenario	Precondition / Test Data	Steps	Expected Result	Expected State	Expected Audit	UAT
TC-RR-001	High	FR-01; UC-RR-001	Create Draft	Valid Requester/site	Create minimum editable Draft	DRAFT; created_by/tenant/site/rowversion/audit correct	DRAFT	Create audit	Yes
TC-RR-002	High	FR-01; File policy	Request attachment	Valid/invalid type; >10MB; PENDING/FAILED/CLEAN	Upload/download	Only allow valid size/type; only CLEAN usable; scoped access	DRAFT	File/access audit	Yes
TC-RR-003	High	FR-02	Submit validation	Missing mandatory/ inactive master	Submit	No transition; controlled validation; input retained	DRAFT	Validation/ correlation	Yes

ID	Pri.	Traceability	Scenario	Precondition / Test Data	Steps	Expected Result	Expected State	Expected Audit	UAT
TC-RR-004	High	BR-14/D-12	Duplicate warning	Matching active request <24h	Submit without then with reason	Warn/block continuation until reason; then SUBMITTED; reason audited	SUBMITTED	Submit + continuation reason	Yes
TC-RR-005	Critical	ST-RR-003	Routing failure	No active approver	Submit/route	Remain SUBMITTED; no auto approve; routing failure event	SUBMITTED	Routing failure audit/ notification	Yes
TC-RR-006	High	UC-RR-003	Approve Request	Valid approver	Approve	APPROVED; decision actor/time/ audit	APPROVED	Decision audit	Yes
TC-RR-007	High	UC-RR-003	Reject Request	UNDER_REVIEW	Reject without/with reason	No reason denied; with reason REJECTED; SLA stopped/breach retained	REJECTED	Decision audit + NTF-REJECTED	Yes
TC-RR-008	High	UC-RR-003	Return for Correction	SUBMITTED/ UNDER_REVIEW	Return without/with reason	No RETURNED state; reason -> DRAFT; edit restored	DRAFT	Decision history + NTF-RETURNED	Yes
TC-RR-009	High	UC-RR-004	Cancel Request	Pre-CONVERTED	Cancel without/with reason	Reason required; CANCELLED; terminal; SLA stop if active	CANCELLED	Cancel audit	Yes
TC-RR-010	Critical	BR-03	Convert	APPROVED	Convert	Exactly one WO OPEN + Request CONVERTED atomically	CONVERTED / OPEN	Convert audit/ outbox	Yes
TC-RR-011	Critical	Concurrency	Conversion retry/failure	Stale/parallel clients	Parallel/retry convert	No duplicate WO; failure leaves APPROVED	APPROVED or CONVERTED	No duplicate audit	No
TC-WO-001	High	FR-09	View/track scope	Users in/out scope	Search/detail/timeline/ file	Authorized visible; out-of-scope denied without leakage	unchanged	Security correlation	Yes
TC-WO-002	High	UC-WO-003	Schedule/Assign	WO OPEN; active team	Schedule	Visit SCHEDULED; WO SCHEDULED; valid times; notification recipients correct	SCHEDULED	Schedule/ assign audit	Yes
TC-WO-003	High	UC-WO-005/006	Reassign/Reschedule	Visit SCHEDULED	Try without/with reason	Reason mandatory; old/new audited; reschedule SCHEDULED via RESCHEDULED	SCHEDULED	Old/new audit	Yes
TC-WO-004	High	UC-WO-007/009	Cancel / MISSED	Before Check-in	Cancel and Mark Missed	Cancel -> CANCELLED; Missed -> MISSED; distinct immutable history	CANCELLED / MISSED	Reason audit	Yes
TC-WO-005	High	UC-WO-008; D-15	MISSED decision	Original MISSED	Reschedule/Follow-up/ Reassign/Cancel decision	Original remains MISSED; new linked SCHEDULED visit for first 3; Cancel creates none; WO unchanged	MISSED + optional new SCHEDULED	Decision/new visit audit	Yes
TC-WO-006	Critical	BR-05/06	Work Session sequence					Session audit	Yes

ID	Pri.	Traceability	Scenario	Precondition / Test Data	Steps	Expected Result	Expected State	Expected Audit	UAT
				Assigned Tech; Visit SCHEDULED	Check-in/Pause/Resume/Check-out	Valid states/times; overlap/invalid denied; audit each action	COMPLETED Visit / CHECKED_OUT		
TC-WO-007	High	Evidence	Check-out evidence	Category/corrective evidence rules	Check-out PENDING/ FAILED then CLEAN	PENDING/FAILED cannot satisfy; CLEAN + summary/outcome required	no transition then COMPLETED	File/work audit	Yes
TC-WO-008	High	UC-WO-020	Submit/review summary	Visit completed	Team Lead submit; Supervisor submit acceptance	IN_PROGRESS -> AWAITING_SUPERVISOR_REVIEW -> AWAITING_CUSTOMER_ACCEPTANCE	AWAITING_CUST OMER_ACCEPTANCE	Summary/ review audit + notification	Yes
TC-WO-009	Critical	BR-07	Customer Accept auth	AWAITING_CUSTOMER_ACCEPTANCE	Other role/contact then designated contact	Others denied; designated Requester/Site Contact -> COMPLETED	COMPLETED	Acceptance round audit	Yes
TC-WO-010	Critical	BR-15	Reject + corrective cycle	AWAITING_CUSTOMER_ACCEPTANCE	Reject/plan/approve/ rework/resubmit/ reaccept	Reason mandatory; exact corrective states; same contact reaccepts; cycle/round trace	COMPLETED after Accept	Full corrective audit	Yes
TC-WO-011	High	BR-08	Close Work Order	WO COMPLETED; cost/ summary reviewed	Non-Supervisor then Supervisor close	Unauthorized denied; Supervisor -> CLOSED; no reopen	CLOSED	Close audit	Yes
TC-WO-012	Critical	BR-09	Cancel WO propagation	Pre-accept; mixed visit states	Supervisor cancel with reason	WO CANCELLED; non-final Visits cancelled; COMPLETED/MISSED preserved; SLA stop	CANCELLED	Atomic propagation audit	Yes
TC-CST-001	High	Cost Summary	Close guard	WO COMPLETED	Try close without/with reviewed cost	Close denied until amount/ currency/prepared/reviewed valid	COMPLETED then CLOSED	Cost review audit	Yes
TC-TIME-001	High	UC-TIME-001	Request correction	Session exists	Supported/ unsupported timestamp; invalid sequence	Only 4 types; old/new/reason stored; invalid sequence rejected	TC PENDING	Request audit	Yes
TC-TIME-002	High	BR-13	Correction decision	TC PENDING	Supervisor approve/ reject	Only Supervisor; reject reason; approved timestamp changes; SLA unchanged	APPROVED/ REJECTED	Decision audit + NTF-TIME	Yes
TC-SLA-001	High	BR-12	SLA lifecycle/breach	Submitted; controlled clock	Advance to terminal	Due=+24h; breach sticky; stop only defined terminals	event-only	SLA event audit	Yes
TC-SLA-002	High	BR-13	SLA Override	Active SLA	Authorized/ unauthorized; before/ after breach	Permission/reason; original/ effective stored; breach never cleared	event-only	Override audit	Yes
TC-SLA-003	High	BR-17	SLA Risk threshold	Active SLA due known			event-only		Yes

ID	Pri.	Traceability	Scenario	Precondition / Test Data	Steps	Expected Result	Expected State	Expected Audit	UAT
					Advance to due-4h; override due; advance again	Risk event at T-4h; recipient matrix; override recalculates future risk; no duplicate event		Risk notification audit	
TC-NTF-001	High	BR-18	Notification failure	Forced provider failure	Execute business transition	Business succeeds; event/recipient FAILED and retryable; no rollback	committed business state	Outbox/retry audit	Yes
TC-NTF-002	High	D-14	Recipient routing	Event set	Trigger each baseline event	Recipients exactly matrix; no cross-site recipient; no sensitive leakage	unchanged	Recipient/delivery trace	Yes
TC-SEC-001	Critical	Security	Cross-tenant/site IDOR	User outside scope	Read/action/download guessed IDs	Denied without sensitive leakage; backend authority	unchanged	Security/correlation	Yes
TC-SEC-002	High	Concurrency	Optimistic concurrency	Two clients same rowversion	Concurrent update/action	First succeeds; stale rejected; no lost update/duplicate transition	one committed result	Conflict audit	No
TC-REP-001	High	FR-09	MVP reports/export	Mixed dataset	Filter/export	Counts/rows correct; scope/PII rules applied	unchanged	Report correlation	Yes
TC-PERF-001	High	NFR Performance	List/Search baseline load	100 concurrent users; 100,000 records/Site; representative filters	Run load against list/search paths	Normal List/Search response <=3 seconds under stated baseline; no scope leakage/data error	unchanged	Performance evidence/log	Yes
TC-UX-001	Medium	NFR Usability	Responsive/browser compatibility	Current Chrome/Edge; desktop/tablet/mobile viewport	Execute critical Requester/Tech/Acceptance flows	No clipped primary actions; validation/loading/empty/error usable; mobile task flow operable	unchanged	Screenshot evidence	Yes
TC-REL-001	High	NFR Reliability	Atomicity/retry	Fault injected between transaction steps	Convert/Reject/Cancel/Close retry/fail	No partial data; retry safe; correlation links audit/outbox	original or fully committed	Transaction/log evidence	Yes
TC-AUD-001	High	Audit	Audit completeness	Representative actions	Execute Reject/Cancel/Reassign/Reschedule/Override/Correction	entity/id/action/actor/time/from/to/reason/old/new/correlation as applicable	unchanged	Audit record itself	Yes

4. UAT Execution Scenarios

UAT ID	Scenario	Business Execution	Acceptance Criteria
UAT-01	End-to-end happy path	Submit -> Approve -> Convert -> Schedule -> Work -> Review -> Accept -> Cost -> Close	WO CLOSED; full audit/evidence/cost/SLA history
UAT-02	Return + resubmit	Return reason -> DRAFT -> edit/resubmit -> approve	No RETURNED state; decision history retained
UAT-03	Duplicate + routing failure	Duplicate warning/reason then route with no approver	D-12 set used; remains SUBMITTED; no bypass

UAT ID	Scenario	Business Execution	Acceptance Criteria
UAT-04	Corrective reacceptance	Reject -> plan -> approve -> corrective Visit -> same contact reaccept	Cycles/rounds traceable; COMPLETED only after Accept
UAT-05	MISSED decision	Mark MISSED then all D-15 decisions	Original remains MISSED; new linked visit only when applicable
UAT-06	Cancel propagation	Cancel WO with scheduled + completed/missed visits	Non-final cancelled; final history preserved; SLA stop
UAT-07	SLA / Risk / Override / Time Correction	Advance to T-4h -> breach -> override -> correction	Risk/breach/override history correct; correction no SLA recalc
UAT-08	Security / attachments	Cross-scope read/action/file attempts	Denied/no leakage; audit/correlation
UAT-09	Notification / reporting	Force delivery failure; run report/export	Business state unaffected; retryable; report scope safe
UAT-10	Performance / Responsive	Baseline load + critical mobile/desktop flows	Performance <=3s; browser/mobile acceptance criteria pass

5. UAT Sign-off Criteria

All UAT=Yes cases passed; Critical/High defect = 0; traceability evidence complete; Security and NFR gates passed; Business Owner/Test Lead sign-off recorded.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อความขัดกัน โดยไม่เปลี่ยน Business Baseline.

6. Performance Test Extension

PERF-DEC-01 - Performance Measurement Statistic (Pending Approval).
Business Baseline ระบุ “Normal List/Search <= 3 seconds at 100 concurrent users and 100,000 records per Site” แต่ไม่ได้ระบุว่า 3 seconds หมายถึง Average, Maximum, p95 หรือ p99. จึง **ห้ามถือ p95 เป็น Requirement ที่อนุมัติแล้ว**. Recommendation สำหรับการอนุมัติคือ p95 end-to-end API response time <= 3.0 seconds; ก่อนอนุมัติ ให้บันทึก Average/p50/p95/p99/throughput/error rate ครบและรายงานผลเป็น diagnostic evidence โดยยังไม่ประกาศ Pass/Fail ของ release gate จาก percentile ไດ percentile หนึ่ง.

PTP-01 - Repeatable Load Test Profile (Technical Test Design)

Item	Technical Test Design - not a new Business Requirement
Baseline concurrency	100 concurrent virtual users - existing baseline.
Dataset	100,000 Repair Request records per Site - existing baseline. Related WO/Visit/Audit/ Notification data must be representative enough to exercise joins and security scope.
Security scope	Use at least 2 tenants and 2 Sites in the test fixture where practical; every request is authenticated. Cross-scope negative traffic is measured separately.
Ramp / duration	Default proposal: 5-minute ramp-up, 30-minute steady-state, 5-minute ramp-down, plus one warm-up run. This may be adjusted in a versioned Test Plan and is not a business requirement.
Filters / sorting / paging	Use representative filters, common sort, first/middle/deeper pages and the same server-side tenant/site scope as production code.
Evidence	Build/commit, environment, DB migration, dataset seed/version, script version, Average/p50/p95/p99, throughput, unexpected error count, API/DB latency, CPU/ memory, DB waits/blocking/slow-query evidence and trace samples.

Performance Test Cases

ID	Priority	Scenario	Expected Result / Evidence
TC-PERF-001	High	Normal List/Search under approved baseline load	Record Average/p50/p95/p99, throughput and unexpected errors. Release Pass/Fail statistic follows PERF-DEC-01 once approved; no scope leakage/data error.
TC-PERF-002	High	Paging/filter/sort including representative deeper pages	Server enforces bounded page input; no unbounded result. Capture SQL/query plan for

ID	Priority	Scenario	Expected Result / Evidence
			critical variations and compare with TC-PERF-001 baseline.
TC-PERF-003	Medium	Report/Export diagnostic	Scope/PII correct; record rows/bytes/duration/memory. No hard latency gate is invented until approved.
TC-PERF-004	Medium	SLA/Outbox/Notification Worker diagnostic	No duplicate business event; capture worker lag/backlog/retry/claim behavior and DB contention; restart remains idempotent.
TC-PERF-005	Medium	File upload/download diagnostic	Authorization correct; only allowed/private file path; record bytes/duration/memory and scan backlog. <=10MB file rule remains unchanged.
TC-PERF-006	High	Concurrent state-changing commands	No lost update/duplicate WO/acceptance/transition; stale requests get controlled conflict; capture lock/wait evidence under concurrent execution.

7. Performance Regression Evidence

Store each approved run with build/commit, DB migration, dataset seed, script version, environment class and result bundle. Full load/stress/soak is not required on every Pull Request; pre-release/UAT performance rerun is required when critical list/query/EF/index/worker/frontend path changes. The allowed regression percentage is **not invented**; compare trends and establish a numeric regression gate only after stable history is approved.